Synopsis

House Listing Model

Problem Id: 5

Student Name: Ashu Khanduri

University Roll number: 2028673

Section: K

Student Name: Srishti Nahar

University Roll number: 2028187

Section: K

Student Name: Anushka Pun

University Roll number: 2028656

Section: K

1. Problem Description:

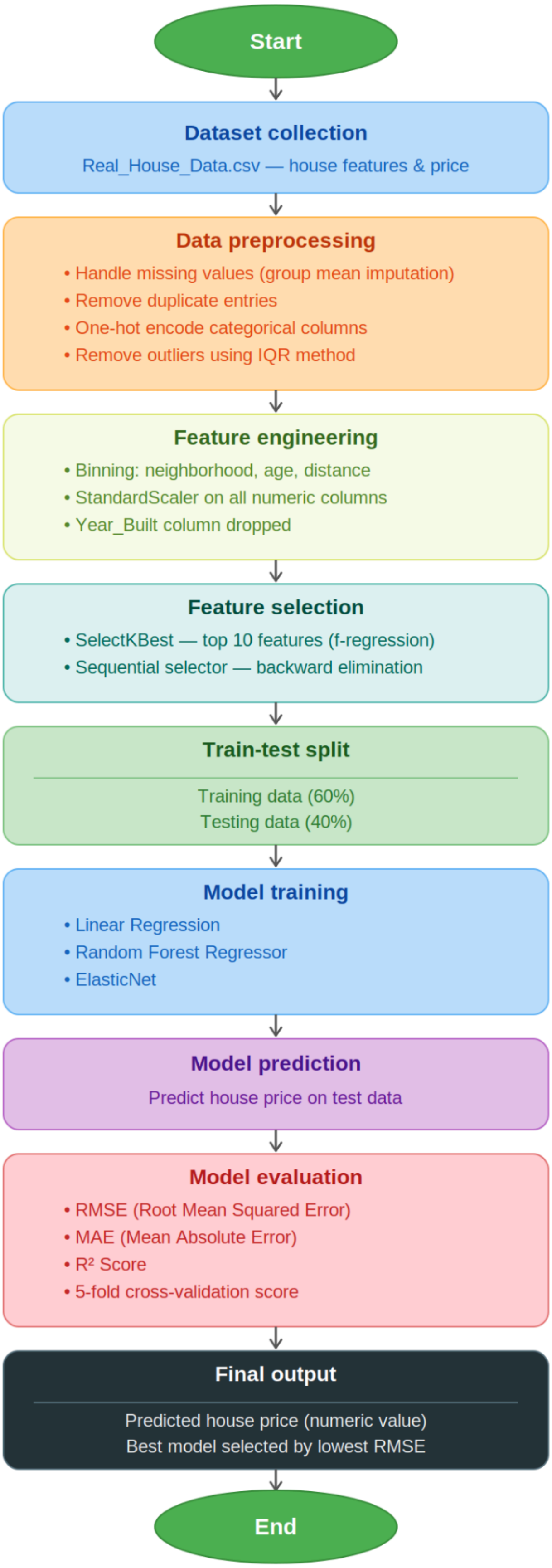
Design and develop a regression-based machine learning system to predict house sale price using a structured real-estate dataset consisting of 25 property-related features. The dataset represents residential housing information collected from different regions and includes both numerical and categorical attributes that influence property valuation.

The dataset contains the following 25 input features: property area (square feet), number of bedrooms, number of bathrooms, number of floors, year built, property age, renovation status, lot size, distance to city center (km), neighborhood quality score, crime rate index, nearby school rating, hospital proximity (km), shopping center proximity (km), public transport accessibility score, parking availability (binary), property type (categorical), construction quality rating, energy efficiency score, water supply reliability score, electricity supply reliability score, internet availability score, green space availability index, flood risk index, and noise pollution level.

The project should begin with Exploratory Data Analysis (EDA) to analyze feature distributions, correlations, and their relationship with the target variable. Perform data preprocessing including handling missing values, encoding categorical features, and normalization or feature scaling.

Develop and train suitable regression models to predict house prices and visualize predicted versus actual values as well as residuals. Finally, evaluate the trained models using appropriate regression performance metrics such as R2 score and error measures such as MAE, MSE, and analyze the impact of different features on the prediction outcome. The project should demonstrate a complete machine learning pipeline from raw data to model evaluation and interpretation.

2. Architecture



3. Models Used

- Linear Regression: The baseline model. It fits a straight-line relationship between selected features and house price by minimizing the sum of squared residuals. Simple, fast, and highly interpretable coefficient values directly show how much each feature influences price. Works best when relationships are approximately linear and multicollinearity is low.
- Random Forest Regressor: An ensemble of decision trees, each trained on a random subset of data and features. Predictions are averaged across all trees, which reduces variance and handles non-linear relationships naturally. It is robust to outliers and requires minimal feature scaling, making it a strong performer on tabular real-estate data.
- ElasticNet: A regularized linear model combining L1 (Lasso) and L2 (Ridge) penalties. L1 shrinks less important feature coefficients to zero (automatic feature selection), while L2 handles correlated features gracefully. Particularly useful here because house features like area, rooms, and quality scores tend to be correlated and ElasticNet prevents any single correlated feature from dominating the model.

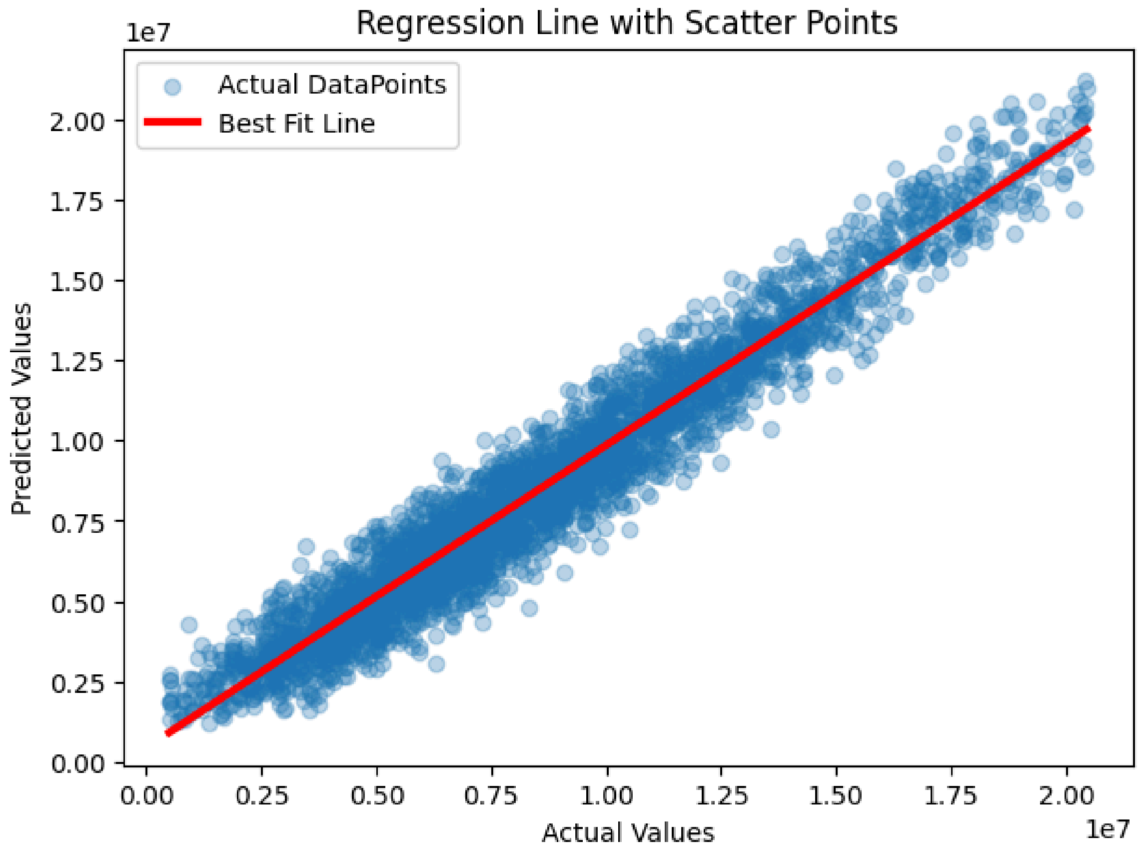
4. Expected Evaluation metrics and Plots to be used:

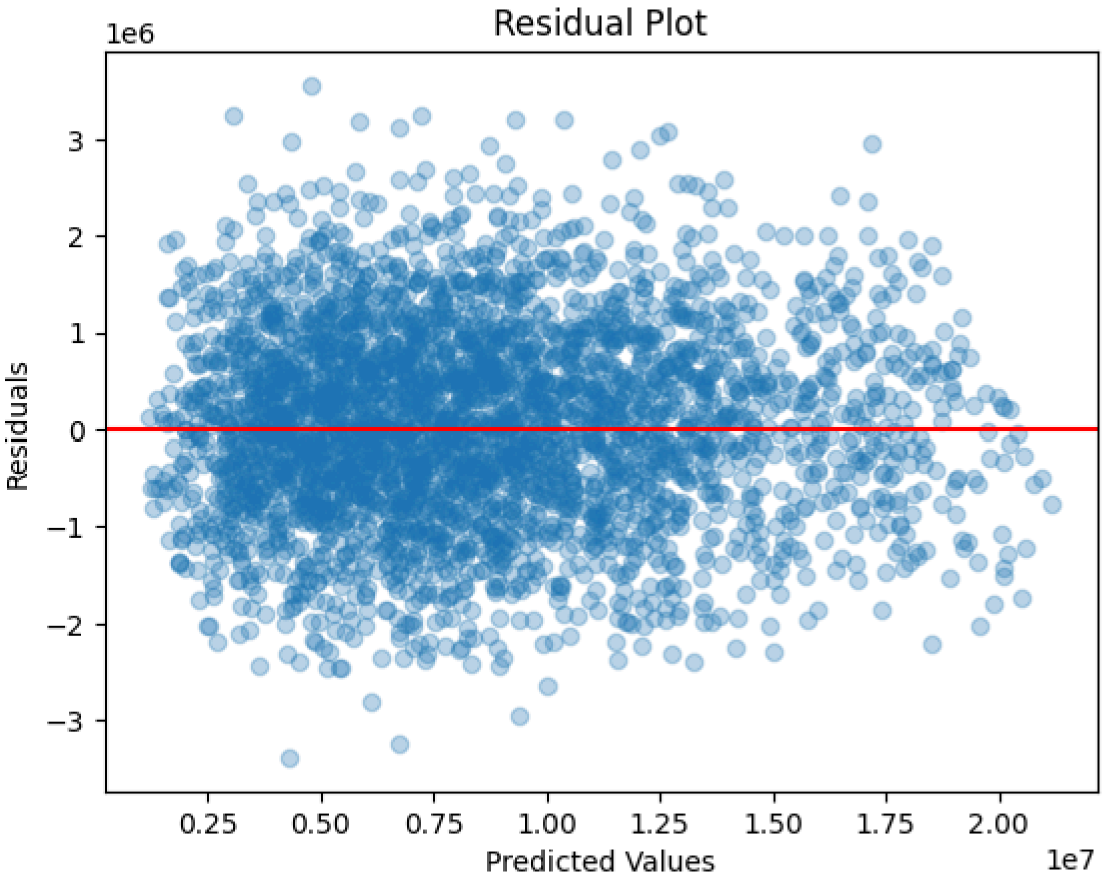
- RMSE (Root Mean Squared Error): It measures how far off the predictions are on average, in the same unit as price. Lower = better. It punishes big errors more than small ones.
- MAE (Mean Absolute Error): It is the average of how much the predictions are off (e.g., "off by ₹5 lakh on average"). It's easier to interpret than RMSE.
- R^2 Score: It tells how well the model explains price variation. $R^2 = 1.0$ is perfect; $R^2 = 0.85$ means the model explains 85% of why prices vary. Higher = better.
- 5-fold Cross-Validation: It splits data into 5 chunks, trains 5 times, and averages R^2 across all runs and checks that the model isn't just memorizing training data (overfitting).

5. Preliminary Results & Visualizations

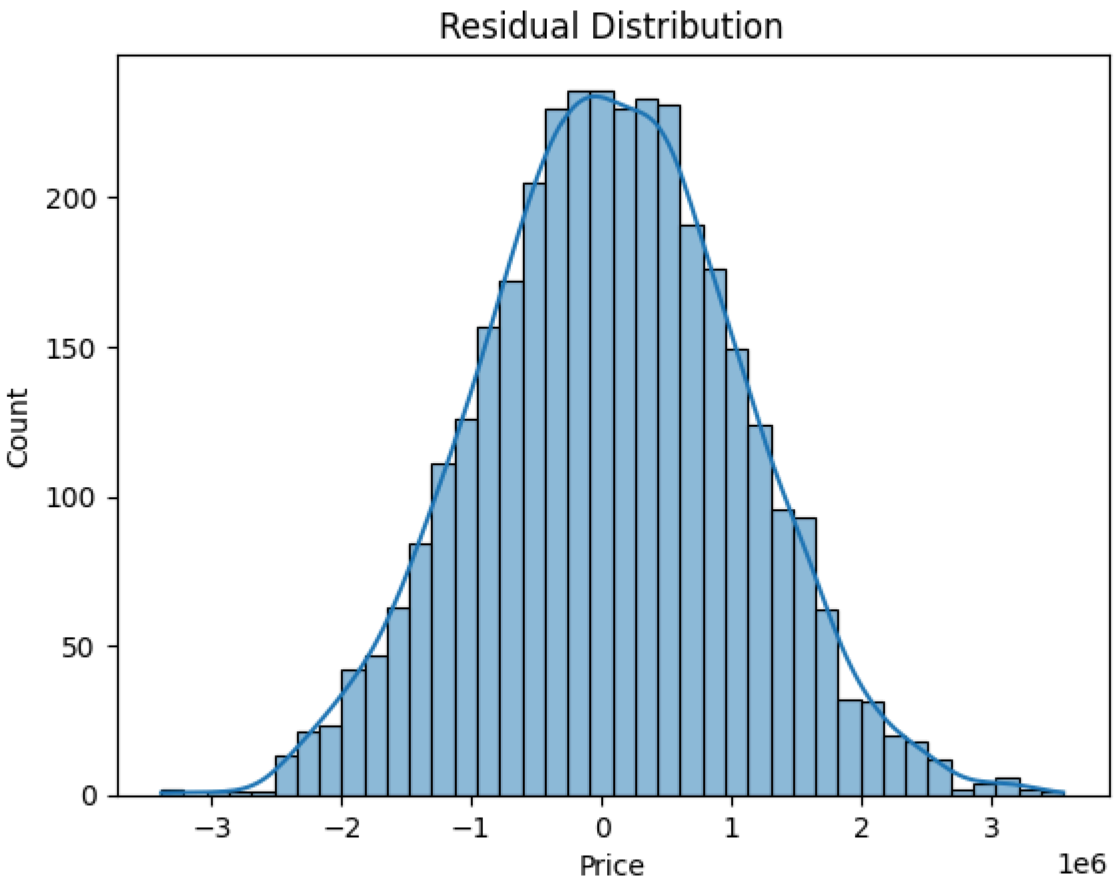
- Price Distribution (Histogram + KDE): Reveals the statistical spread of target values. A right-skewed distribution (common in real estate) indicates that a few high-value properties dominate the range, justifying outlier removal prior to modeling.
- Scatter Plots (Area / Distance / Crime vs Price): Exploratory analysis confirming expected feature-price relationships: larger property area correlates positively with price, while greater distance from the city center and higher crime index correlate negatively. This validates the relevance of these features before model training.
- Boxplot of Price: Identifies statistical outliers using the IQR method. Extreme values are flagged and removed to prevent them from distorting regression coefficients and inflating error metrics.
- Actual vs Predicted Scatter (with Best Fit Line): A key diagnostic post-training. Points clustering tightly along the diagonal best-fit line indicate low prediction error. Systematic deviation suggests model bias or unaccounted non-linearity.

- Residual Plot (Predicted vs Residuals): Tests a core linear regression assumption that errors are randomly and evenly distributed. A fan-shaped or curved pattern would indicate heteroscedasticity or a need for non-linear modeling.
- Residual Distribution (Histogram + KDE): Validates the normality of errors, a prerequisite for reliable inference in linear models. A bell-shaped curve centered at zero confirms well-calibrated predictions.





Residual Plot
1e6
3
2
1
0
−1
−2
−3
Residuals
0.25
0.50
0.75
1.00
1.25
1.50
1.75
2.00
Predicted Values
1e7



Residual Distribution
200
150
100
50
0
Count
−3
−2
−1
0
1
2
3
Price
1e6

6. References

- Platform Required: jupyter notebook
- Books and Link Sources:

1. https://www.geeksforgeeks.org/ [understanding theory and concepts]

2. Chatgpt/Gemini : For helping in understanding what to do and why we are doing it.

3. Pandas Official Documentation – https://pandas.pydata.org/docs/

4. Scikit-learn Official Documentation- https://scikit-learn.org/docs/